

Supra AI

AI-Assisted Supplier Compliance Screening for Consumer Electronics Retail

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Industry: Omnichannel Consumer Electronics Retail · Company size: Mid-sized (~200 employees)

Supplier compliance is becoming a growth constraint

Market is scaling fast

Compliance software is projected to roughly double this decade — from about \$35.8B in 2025 to \$78.9B by 2033, growing over 10% a year.

The stakes keep rising

GDPR penalties reach up to €20M or 4% of global revenue. The EU AI Act's obligations are phasing in through 2025–2027 — this isn't a one-time compliance push, it's a moving target.

The workload is compounding

The average company now manages roughly 286 third-party supplier relationships, up from 237 just a year earlier — each one generating its own paperwork trail. Manual review doesn't scale at that growth rate.

We kept the industry and use case — here's why

What was validated

- ✓ 12/12 (100%) agreement with labelled ground truth
— 7/7 synthetic + 5/5 real-world certificates
- ✓ Deterministic screening can be cleanly separated from AI extraction
- ✓ Real-world documents surfaced failure modes synthetic data alone would not expose
- ✓ A narrow, human-in-the-loop workflow gives Chleo a story that isn't a black box

What MVP strengthens

- LangGraph-based agentic extraction pipeline replacing the lighter Round 1 pipeline
- Expanded real-world benchmark: 13/13 audit decisions, 95.9% overall extraction accuracy
- Supplier Gap Notice generation and a reviewer-facing interface
- Audit Copilot Chat — audit-scoped, evidence-grounded reviewer assistant
- ROI, EU AI Act, GDPR, and pilot-based strategic deployment plan

Compliance review doesn't scale with the catalogue

“Imagine a 200-person European electronics retailer with 2,000 active SKUs. CE certificates, RoHS test reports, and declarations of conformity arrive by email and PDF — and a compliance reviewer still has to read, extract, and cross-check every one by hand.”

1

Fragmented documentation

Certificates, spreadsheets and email threads sit across suppliers, products and repositories with no single source of truth.

2

Gaps are hard to catch early

Missing, expired, or inconsistent evidence is easy to miss until a supplier or product is already at risk.

3

Manual checks don't scale

Standardising review effort gets harder every time the catalogue or the supplier base grows.

4

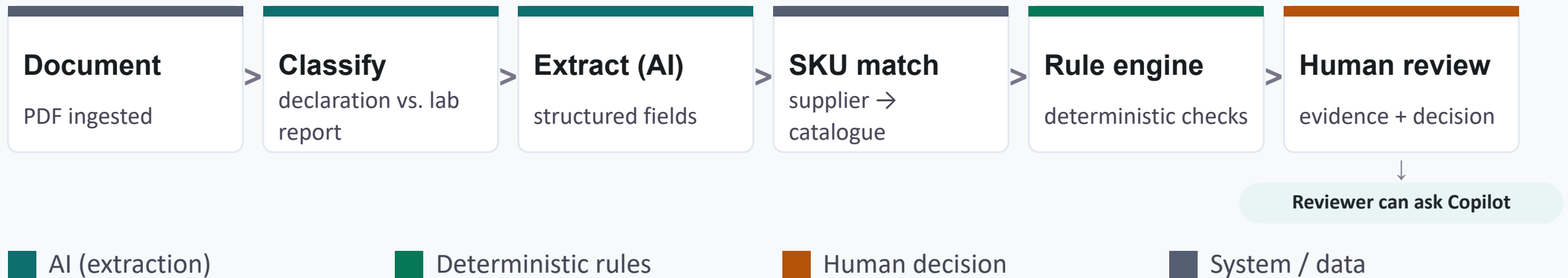
AI needs to be explainable

Chleo's core fear is a black box — any AI assistance must show its evidence, not just a verdict.

AI extracts, deterministic rules decide, humans approve

Supra AI turns unstructured supplier compliance documents into structured, evidence-backed screening results — flagged for human review, never auto-decided.

End-to-end workflow



The system type

Human-in-the-loop AI document intelligence and compliance screening. The LLM extracts and explains; a deterministic, testable rule engine screens; a qualified reviewer makes the final compliance call. Copilot Chat helps the reviewer investigate a decision after the fact — it never makes one. APPROVED / FLAGGED / REJECTED are workflow states, not legally binding regulatory decisions.

The n8n proof of concept

What the POC shows

- 1 Trigger:** A supplier compliance PDF (Declaration of Conformity or lab report) is submitted into the workflow.
- 2 AI extraction:** The LLM step reads the document and returns structured fields — supplier, product, standards, dates, certificate reference.
- 3 Rule screening:** Deterministic checks run against expiry, mandatory standards, and the SKU catalogue.
- 4 Output:** A prioritised, evidence-linked finding is written out — with the source text attached, not just a verdict.

Tools used

- n8n — workflow orchestration
- LLM API — document extraction
- LangSmith — tracing & evaluation
- Python + pandas — data prep

Proves / doesn't prove

Proves: end-to-end feasibility — extraction → rules → flagged output.

Doesn't prove: production-scale reliability or generalisation across every document type.

The ROI case

274%

12-month ROI
base case

781%

36-month ROI
base case

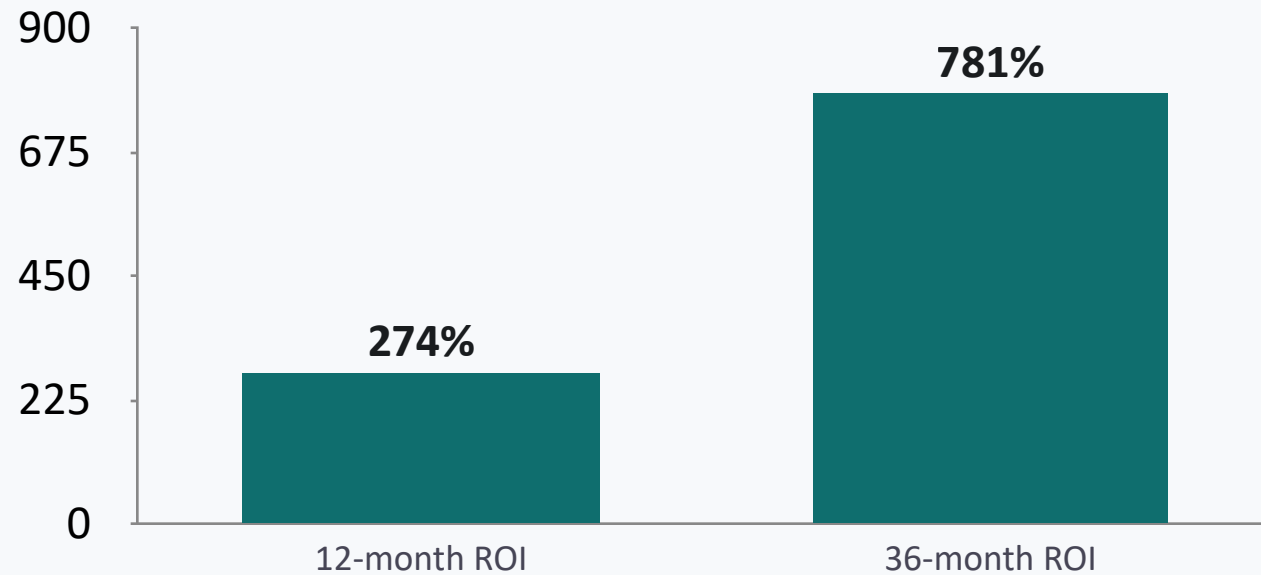
~3 mo

Break-even
base case

€18,000

Annual value
reviewer time released

ROI by scenario



Base-case assumptions

Documents / year	1,800 (150/mo)
Manual review time	30 min / document
Time saved with Supra AI	40%
Reviewer loaded cost	€50 / hour
Upfront implementation	€4,150
Monthly tooling	€55
AI inference cost	~\$0.0004 / document

These are scenario-based planning estimates, not confirmed customer savings — to be recalculated during the pilot.

The three risks most likely to derail this project

Technical

Poor-quality or scanned PDFs

MITIGATION

Document-quality detection, unsupported-format handling, and human escalation for low-confidence scans.

Technical

Incorrect AI extraction

MITIGATION

Labelled evaluation data, field-level accuracy checks, evidence-linked extraction, and mandatory human review.

Operational

Incorrect SKU matching

MITIGATION

Exact + normalised matching, ambiguity detection, human review for uncertain matches, source values retained.

These risks are manageable — every one of them is caught by the same safety net: evidence-linked findings plus mandatory human review before any decision is acted on.

How compliant is this?

EU AI Act

Limited / minimal risk

Why: Supra AI does not fall under an Annex III high-risk category — it doesn't autonomously decide safety, employment, credit, or law-enforcement outcomes. It supports document review; every compliance status stays provisional until a qualified human approves it.

Obligations addressed:

human oversight (100% of final decisions human-approved), transparency that outputs are AI-generated, and traceable run logs via LangSmith as a record-keeping baseline.

Copilot Chat:

same constraints apply — read-only, evidence-grounded, logged to the same LangSmith trail. It explains a decision; it cannot change one.

Gap:

classification is our own reasoned assessment, not a formal legal opinion — confirming it with counsel is a Phase-1 exit item.

GDPR

Data-minimised

Personal data touched: minimal — mainly supplier/company and product data; occasional signatory names on certificates. No customer personal data. Capstone testing uses synthetic and public documents only.

Legal basis:

legitimate interest — verifying supplier compliance documentation for product safety and regulatory conformity.

DPIA:

recommended before processing real supplier documents at scale, given the (limited) presence of personal identifiers on certificates.

Working MVP → controlled pilot → production

01

Working MVP

NOW

Core screening workflow runs end to end on the validation set; benchmark held as the regression baseline.



02

Controlled pilot

~8 WEEKS

One customer, one product category, real documents. Measure review-time reduction, critical misses, evidence quality.



03

Production

POST-PILOT

Validated workflow becomes part of the customer's normal supplier-review process, with agreed controls.

Go-to-market, in three sentences

Sold to mid-sized European omnichannel retailers with compliance-heavy electronics catalogues. Entry via a paid discovery pilot, converting to a production subscription once measured value is confirmed. Positioned as an auditable AI-assisted screening layer — not autonomous compliance decision-making.

What greenlights the next phase

- ✓ Measured review-time reduction vs. manual baseline
- ✓ Zero unresolved critical misses on the validation set
- ✓ Reviewers can understand and trust every finding

From POC to the beginning of the real product

What the POC showed was possible

- Single-pass LLM extraction on clean PDFs
- Rule screening on a small synthetic test set
- Basic LangSmith tracing for one pipeline run

What the MVP actually runs

- ✓ LangGraph agentic extraction pipeline (graph.py, schemas.py)
- ✓ SKU/MPN resolution + deterministic rule engine at scale
- ✓ Supplier Gap Notice generation for flagged/rejected cases
- ✓ Audit Copilot Chat — reviewer asks, evidence answers

Validated benchmark results

13/13

audit decisions agree with ground truth

95.9%

overall field-level extraction accuracy

100%

extraction on laboratory test reports

CONCLUSION

A pilot is the next step — not a leap of faith

Business value

Supra AI cuts supplier-document screening effort while keeping every compliance decision human-approved.

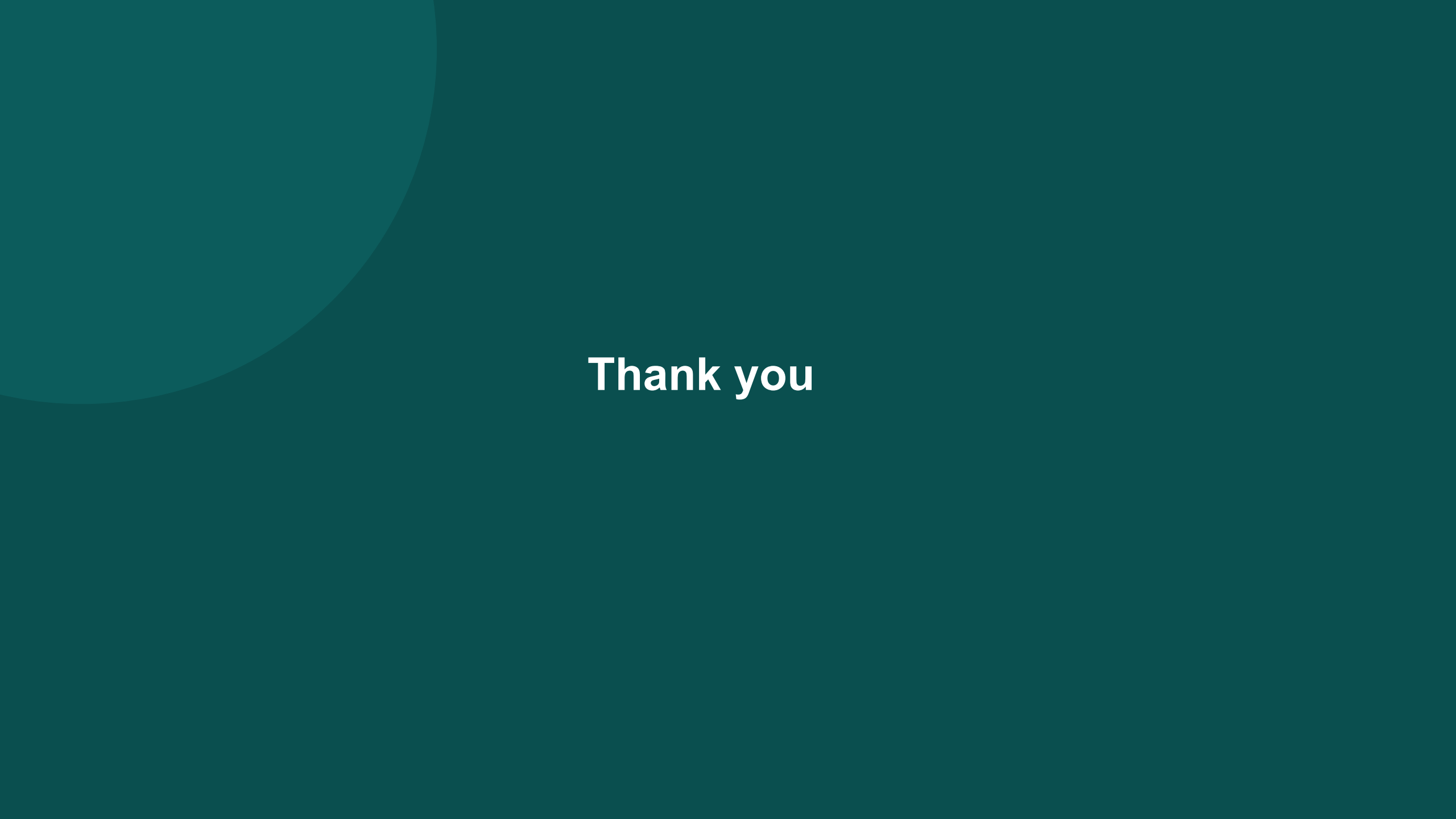
Compliance status

Positioned for limited / minimal-risk deployment with data-minimised handling — pending formal legal sign-off.

Evidence so far

13/13 audit decisions and 95.9% extraction accuracy on the current benchmark

Next step: a 8-weeks pilot on Chleo's consumer-electronics catalogue to validate these numbers against real documents.

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Thank you